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Analysis and Detection of LLM-Generated Text in Swedish

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Abstract

This thesis investigates investigate systematic differences between text generated by large language models (LLMs) and human-written text in the Swedish language. In addition, the thesis analyses significant differences in the ability to detect LLM-generated Swedish text between humans and AI-detection tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

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Preface

I would like to thank the following people...

1 Introduction

The invention of transformer-based Large Language Models (LLMs) has created a new wave of textual data that is entirely generated by machines and very closely emulates, to the naked eye, text written by humans. The study of the linguistic properties of LLM-generated text has emerged from this recent phenomenon as a growing topic of natural language processing (NLP). This topic differs in focus from that focus primarily on the results of training AI-detection classifiers; rather, the topic focuses on how to utilise pre-existing analysis methods of to analyse the textual characteristics and distributions of LLM-generated text. Recent research has focused primarily on English-language textual analysis as the dominant type of data analysed. However, there has been much less research on the characteristics of LLM-generated text in languages other than English. Hence, there is a lot of opportunity for exploration of this topic in many other languages.

Analysing the characteristics of LLM-generated data arguably enhances the ability to shed light on the mechanisms of LLMs, potentially contributing to transparency and LLM-probing research within the NLP space. As these tools continue to generate unbelievable amounts of textual data at a break-neck speed (« insert some figure taken recently on number of TB per year »), changing the landscape of data proliferated through the internet and in other published textual sources, such studies could also shed light on what kind of text is being read by the literate population, which styles are dominating, and what kind of language structures and content is being prioritised through the usage of these generative AI.

In addition, studies such as these can inform readers of potential heuristics to use when reading a text, influencing their intuition to scrutinise a text and potentially infer their origin from a machine. The artificial intelligence detection (AI-detection) space can uncover instances where an AI-tool has been used to generate text. The AI-detection process is an important topic of study, as these methods can mitigate the risk of bad agents influencing the text of the internet within politics, as well as in the educational space, where the learning process of a student has been compromised by plagiarism through the usage of a generative tool.

1.1 Research questions

The intended purpose of this study is to investigate systematic differences (if any) between text generated by large language models (LLMs) and text written by humans in the Swedish language. In addition, the study will investigate whether there are any significant differences in the ability to detect LLM-generated Swedish text between humans and LLM tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

1. Are there systematic differences between LLM-generated and human-written formal text in Swedish?
2. Are there systematic differences between LLM-generated and human-written informal text in Swedish?

3. Are there differences between humans and AI-based detection tools in their ability to detect LLM-generated Swedish text?

1.2 Experimental-set up of the thesis

The thesis is comprised of two experiments: 1) textual analysis, and 2) LLM detection. The first experiment focuses on how human-written and LLM-generated text differ in surface-level and content-level characteristics. There are two domains of text used: 1) a formal domain comprised of Bachelor's level thesis abstracts, and 2) an informal domain of internet thread comments. The LLM used to generate the texts is OpenAI's GPT-5.2. The second experiment focuses on the differences in ability and thinking process between humans and LLMs in detecting LLM-generated text. The LLM of focus is DeepSeek (? plus model version), and the accuracy and reasoning process will be evaluated for both Swedish-native speakers and this LLM in detecting both the formal and informal data.

2 Background

This chapter will provide an overview of previous studies related to this topic. First, it will outline the development of large language models and their text generation abilities for English and Swedish. Second, it will describe previous research comparing AI-generated text with human-written text. The chapter will then outline tools and human participant studies related to AI-generated text detection.

2.1 Large Language Models and text generation

2.1.1 Large Language Model architectural development

At the time of writing this thesis, the dominant underlying architecture of LLMs is the transformer architecture, first described in [vaswaniz2017attention](#). In this study, the authors designed a basic neural network architecture with a multi-head attention mechanism and no recurrent unit structure. The transformer architecture has demonstrated superior performance and decreased training time compared to its recurrent neural network predecessors. The strength of transformers lies in their ability to handle long dependencies between different elements of input sequences, meaning that it can calculate how relevant different parts of input are to one another, no matter how far apart one element is from another within a sequence. As a result, transformer networks can be adapted to a variety of tasks, including optical recognition, speech processing, and text generation ([islam2023](#)).

Specifically look at LLM architecture and text generation itself, its relationship with training on data.

2.1.2 LLMs for the English language

English has a vast amount of data - list major datasets used for LLM development in English.

Talk about compositions of English-dominant LLMs - data type

2.1.3 LLMs for the Swedish language

Ohman et al. (2023) curated a 1.2 terabyte dataset of text in all Scandinavian languages in addition to high-quality English data, called the Nordic Pile dataset. Holmstrom et al. (2023) initiated an investigation of the extent to which pre-training GPT-3 LLMs on Swedish only is advantageous over leveraging pre-trained knowledge (from English). It also examined whether English-based pre-trained models demonstrated unwanted transfer of Anglophone cultural aspects over to the Swedish context. The findings of Holmstrom et al. (2023) demonstrated that LLMs exposed to limited Swedish during training can be highly capable and transfer competencies from English "off-the-shelf", including mathematical reasoning and Swedish cultural adaptation. In addition, the study showed that GPT-3 models did not appear to exhibit unwanted over-generalisation of cultural aspects to Swedish.

2.2 Textual comparison of LLM-generated and human-written text

Several studies have already done that analyse the linguistic characteristics of LLM-generated text in English.

2.2.1 Linguistic properties of LLM-generated text

Several studies have already done that analyse the linguistic characteristics of LLM-generated text in English.

Munoz-Ortiz et al. (2024) compared the linguistic components of human and LLM-generated news articles and compared their patterns among six generative LLMs. For their prompting strategy, they asked the models to generate a news paragraph based on the news article headline its first words. For their linguistic analysis, they compared the human-written and LLM-generated text groups through differences in their linguistic structures, including vocabulary distribution, part-of-speech tag distributions, dependency arc lengths, constituent lengths, cosine similarity, and emotional category tagging. The results of this analysis showed significant differences between the human-written and LLM-generated text on all textual analysis metrics. At the lexical level, 5 of the 6 LLM models outputted text with a more restricted vocabulary than the human-written text. For semantics, the models were found to have less of an inclination to display aggressive negative emotions (such as fear or anger) than humans.

Liao et al. (2023) analysed the differences between human-written and ChatGPT-generated text within the medical domain. Medical texts were compared by their vocabulary distribution, part-of-speech tag distribution, dependency arcs, sentiment analysis, and perplexity. In addition, they implemented machine learning methods to detect ChatGPT-generated medical text. The results of the study found that human-written medical texts contained more concrete information, had higher vocabulary diversity, and contained information that human experts would consider more useful. The ChatGPT-generated medical texts appeared more fluent and logical, but expressed more generalised terminology rather than specific, effective information for the problem's context. Overall, this would point to the human-written medical text being more tailored to the situational modelling of the medical case (situational modelling defined in Mahowald et al., 2024) rather than the LLM-generated text.

- discuss differences in text stylistic

2.2.2 Other properties of LLM-generated text

Previous work has also shown evidence of a distinction between the reasoning patterns of LLM tools when compared to those of humans. (CITE) evaluated the medical reasoning capabilities of DeepSeek R1, computing its outputs to the reasoning patterns of medical domain human experts. The study performed qualitative and quantitative analyses of AI- and human-medical reasoning for 100 clinical cases from the MedQA dataset. The study found a 93% diagnostic accuracy for DeepSeek R1's performance. Many of the LLM's errors were attributed to over-focus on the details of a medical problem, rather than a more holistic and intuitive (?) understanding by humans, as well as an insufficient understanding of medical protocols when compared to human experts. The study also found a strong correlation between the length of output and

the error rate of LLMs; that is, results from LLMs that were longer were more likely to contain incorrect diagnoses or medical details.

- discuss differences in text stylistic

2.3 AI-generated text detection studies

2.3.1 AI-detection tool development

2.3.2 Human participant AI-detection studies

Landberg (2024) investigated the effectiveness of AI chatbots to detect texts created by AI in Swedish and English. The main component of the study A recent study investigated LLM reasoning in Swedish. Moell and Boye (2025) studied the performance of LLMs on language complexity measurement tasks (LIX readability metric and Average Dependency Distance). The results of the study found that all models demonstrated at least some capacity for each of these tasks, but that LLMs remain prone to errors in explicit calculations and syntactic analysis.

2.4 Background summary

3 Methodology

This chapter details the methodology of this thesis, of which there are several components. The methodology is comprised of two main parts: 1) a comparison of textual elements between human-written Swedish and LLM-generated Swedish, and 2) a comparison between the detection rates and strategies of LLM-generated Swedish. For both main parts, this chapter details the input data, tools and evaluation methods in their experimental set-ups.

3.1 Data collection

3.1.1 Human-written formal data

The formal human-written dataset comprises of (n=insert) samples of Bachelor's level thesis abstracts and keywords (could be 200+ if you wanted to) sourced from the Digitala Vetenskapliga Arkivet (DiVA) portal. All Swedish abstracts and keywords were written in Swedish. Where the abstract was Swedish but the keywords were in English, the English keywords were translated into their Swedish equivalents. The date range of these abstracts was 2007-2017 in order to mitigate the chance of any transformer-based generative models influencing the creation of these abstracts. The topics of the thesis abstracts spanned across several topics (Humanities, Physical Sciences, Medical Studies, Agriculture etc.).

3.1.2 Human-written informal data

The informal human-written dataset comprises of (n=insert) Swedish-language Reddit questions, with (n=insert) replies to each question, all scraped from r/Sweden using the PushShift and Arctic Shift APIs. To mitigate the possibility of transformer-based generative AI tools being involved in the creation of the subreddit's questions and replies, only questions and comments with a timestamp before January 2017 (2017-01-01) were scraped. To ensure that the post's titles were in question form and in Swedish, the scraper script checked for the presence of a question mark ("?",) and used the <langdetect> Python library to identify the post as being written in Swedish ("sv").

3.1.3 LLM-generated formal data

The formal human-written dataset comprises of an LLM-generated version of each human-written sample of the DiVA Bachelor's-level thesis abstracts. The model used was OpenAI's GPT 5.2 model (<gpt-5.2>). The prompt used as input to generate the comments was:

"Titta på titeln och nyckelorden och skapa en sammanfattning i kandidatuppsatsstil med dina egna ord: <title, keywords>."

English translation: Look at the title and keywords and create a summary of the thesis abstract in your own words: <title, keywords>

3.1.4 LLM-generated informal data

3.2 Experiment 1: textual analysis

The aim of the textual analysis experiments is to investigate meaningful differences between LLM-generated and human-written Swedish. This method is based on the hypothesis that there are indeed meaningful differences in the writing style between the two textual sources in Swedish. The method aims to cover both surface-level analyses of the text, that is, investigating structural differences between the word forms and surface-level composition between the two modes; as well as content-level analysis, that is, sentiment analysis, entities, and themes of the texts.

3.2.1 Surface-level analysis

For the surface-level analysis, the spaCy package was used (`sv_core_news_sm`) to collect distributions of: vocabulary distribution, part-of-speech distribution, dependency analysis (sprakbanken?), dependency arch length, average word length, average sentence length.

3.2.2 Content-level analysis

For the content analysis, the study used the sentiment analysis, named entity recognition, and thematic clustering(?) techniques to uncover recurring ideas and themes within both the abstracts and comments. Hypothesis: this is going to be more important for the Reddit comments than the abstracts.

3.3 Experiment 2: LLM-generated text detection

3.3.1 AI detection

Another LLM (DeepSeek or whatever) to do AI detection - quantitative measure (accuracy), qualitative measure (reasoning)

3.3.2 Human detection

An LLM-detection game (AI eller inte, AI or not in English) was developed as a web app

3.3.3 LLM-detection experimental set-up

There are two main types of textual analysis for this data: 1. surface-level linguistic analysis that investigates and tries to generalise what the differences are in linguistic formation between human and AI Swedish; and 2. ontological (?) analysis that investigates the CONTENT and information tendencies of the human Swedish compared to the AI Swedish. The method of the surface-level linguistic analysis will follow what has been done before in previous studies - there is one that analyses differences in medical abstracts in English (li2023).

Just ONE Large Language model – GPT-5

Two types of data: formal (Bachelor's thesis abstracts) and informal (Reddit comments). We used one model, GPT-5, to generate the synthetic data for both the formal and informal data. Another LLM (DeepSeek or whatever) to do AI detection - quantitative measure (accuracy), qualitative measure (reasoning)

4 Results

text here

5 Discussion

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6 Conclusion

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